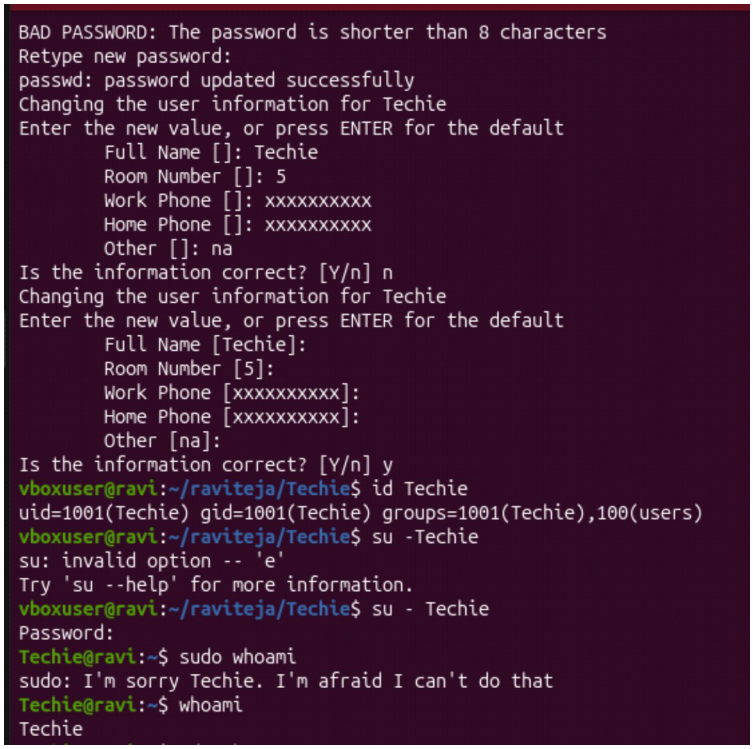
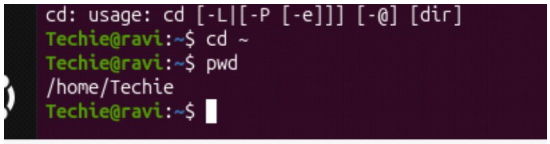
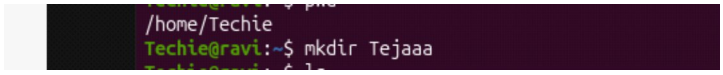
Create user with name Techie and provide sudo access to user.



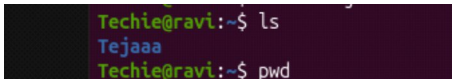
Navigate to the home directory.



Create a new directory.



List the contents of a directory.



Change the current directory.

```
Tejaaa
Techie@ravi:~$ pwd
/home/Techie
Techie@ravi:~$ cd Tejaaa
```

Create a new empty file.

```
Techie@ravi:~/Tejaaa$ touch Rajaa
```

View the contents of a file.

```
Techie@ravi:~/Tejaaa$ vi Rajaa
```

Copy a file to another location.

```
Techie@ravi:~/Tejaaa$ touch krish
Techie@ravi:~/Tejaaa$ cp Rajaa krish
Techie@ravi:~/Tejaaa$ cat krish
B
B


B

iiiiidd
i
ddIC
HELLO WO
```

Move a file to another location.

```
Techie@ravi:~/Tejaaa$ touch vams
Techie@ravi:~/Tejaaa$ mv krish vams
Techie@ravi:~/Tejaaa$ cat vams
B
B

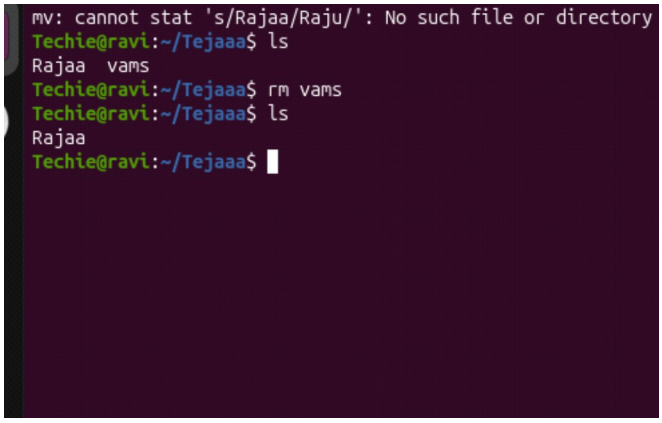

B

iiiiidd
i
ddIC
HELLO WO
```

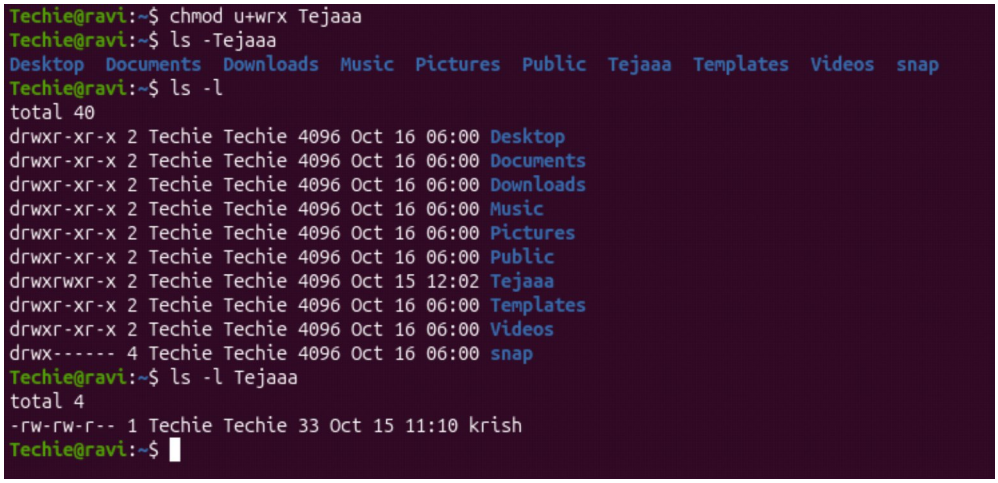
Rename a file.

```
Techie@ravi:~/Tejaaa$ mv Rajaa krish
Techie@ravi:~/Tejaaa$ ls
krish
Techie@ravi:~/Tejaaa$
```

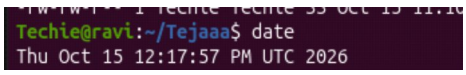
Delete a file.



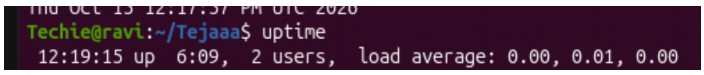
Grant or revoke permissions on a file or directory.



View the current date and time.



Check the system uptime.



View the running processes.

```
Techie@ravi:~/Tejaaa$ ps
    PID TTY          TIME CMD
  13853 pts/1    00:00:00 bash
  14942 pts/1    00:00:00 ps
Techie@ravi:~/Tejaaa$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 06:10 ?        00:00:23 /usr/lib/systemd/systemd --system --deserialize=65
root           2       0  0 06:10 ?        00:00:00 [kthreadd]
root           3       2  0 06:10 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 06:10 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 06:10 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 06:10 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 06:10 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 06:10 ?        00:00:00 [kworker/R-netns]
root          10       2  0 06:10 ?        00:00:00 [kworker/0:0H-kblockd]
root          13       2  0 06:10 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 06:10 ?        00:00:01 [ksoftirqd/0]
root          15       2  0 06:10 ?        00:00:00 [rcu_preempt]
root          16       2  0 06:10 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          17       2  0 06:10 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          18       2  0 06:10 ?        00:00:00 [migration/0]
root          19       2  0 06:10 ?        00:00:00 [kprobe-optimizer]
root          20       2  0 06:10 ?        00:00:00 [idle_inject/0]
root          21       2  0 06:10 ?        00:00:00 [cpuhp/0]
root          22       2  0 06:10 ?        00:00:00 [kdevtmpfs]
root          23       2  0 06:10 ?        00:00:00 [kworker/R-inet_frag_wq]
root          24       2  0 06:10 ?        00:00:00 [rcu_tasks_kthread]
root          25       2  0 06:10 ?        00:00:00 [rcu_tasks_rude_kthread]
root          26       2  0 06:10 ?        00:00:00 [kauditd]
root          27       2  0 06:10 ?        00:00:00 [khungtaskd]
root          28       2  0 06:10 ?        00:00:00 [oom_reaper]
root          31       2  0 06:10 ?        00:00:00 [kworker/R-writeback]
root          32       2  0 06:10 ?        00:00:02 [kcompactd0]
```

```
vboxuser    3188    2434  0 06:11 ?        00:00:00 /usr/libexec/gvfsd-trash --spawner :1.21 /org/gtk/gvfs/exec_spaw/0
vboxuser    3282    2227  0 06:11 ?        00:00:02 /snap/snapd-desktop-integration/391/usr/bin/snapd-desktop-integration
vboxuser    3356    2189  0 06:11 ?        00:00:00 /usr/libexec/xdg-desktop-portal-gtk
vboxuser    3495    2189  0 06:11 ?        00:00:39 /usr/bin/ptyxis --gapplication-service
vboxuser    3502    3495  0 06:11 ?        00:00:02 /usr/libexec/ptyxis-agent --socket-fd=3 --rlimit-nofile=1024
vboxuser    3519    2479  0 06:11 ?        00:00:00 /usr/bin/Xwayland :0 -rootless -noreset -accessx -core -auth /run/user/1000/.mutter-Xwaylandauth.O
vboxuser    3524    2189  0 06:11 ?        00:00:00 /usr/libexec/gsd-xsettings
vboxuser    3529    2479  0 06:11 ?        00:00:00 /usr/libexec/mutter-x11-frames
vboxuser    3550    3502  0 06:11 pts/0    00:00:00 /usr/bin/bash
vboxuser    3560    2189  0 06:11 ?        00:00:00 /usr/libexec/ibus-x11
root        3590       1  0 06:12 ?        00:00:03 /usr/libexec/fwupd/fwupd
root        5419    3550  0 06:24 pts/0    00:00:00 sudo su root
root        5428    5419  0 06:25 pts/1    00:00:00 sudo su root
root        5429    5428  0 06:25 pts/1    00:00:00 su root
root        5434       1  0 06:25 ?        00:00:01 /usr/lib/systemd/systemd --user --deserialize=22
root        5436    5434  0 06:25 ?        00:00:00 (sd-pam)
root        5457    5434  0 06:25 ?        00:00:00 /usr/bin/dbus-daemon --session --address=systemd: --nofork --nopidfile --systemd-activation --sysl
root        5459    5429  0 06:25 pts/1    00:00:00 bash
root        5460    5434  0 06:25 ?        00:00:00 /snap/snapd-desktop-integration/391/usr/bin/user-session-helper /snap/snapd-desktop-integration/39
root        5494    5434  0 06:25 ?        00:00:00 /usr/libexec/xdg-document-portal
root        5498    5434  0 06:25 ?        00:00:00 /usr/libexec/xdg-permission-store
root        5507    5494  0 06:25 ?        00:00:00 fusermount3 --auto-unmount -- /run/user/0/doc
root        9768    5459  0 06:36 pts/1    00:00:00 su vboxuser
vboxuser    9770    9768  0 06:36 pts/1    00:00:00 bash
root       12004       2  0 06:37 ?        00:00:00 [psimon]
root       12005       2  0 06:37 ?        00:00:00 [psimon]
root       12255       2  0 06:52 ?        00:00:00 [kworker/u5:1-ttm]
root       13809       2  0 10:39 ?        00:00:01 [kworker/u4:1-events_unbound]
root       13849    9770  0 10:45 pts/1    00:00:00 su - Techie
Techie     13853   13849  0 10:45 pts/1    00:00:00 -bash
root       14075       2  0 11:10 ?        00:00:00 [kworker/u4:0-events_unbound]
root       14076       2  0 11:10 ?        00:00:02 [kworker/0:2-events]
root       14701       2  0 12:00 ?        00:00:00 [kworker/u4:4-ext4-rsv-conversion]
root       14772       2  0 12:10 ?        00:00:00 [kworker/0:1-cgroup_release]
root       14774       2  0 12:10 ?        00:00:00 [kworker/u4:2-events_power_efficient]
vboxuser   14817    2479  0 12:12 ?        00:00:00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P /usr/share/gnome-shell
root       14911       2  0 12:17 ?        00:00:00 [kworker/u4:3-events_unbound]
Techie     14952   13853  0 12:21 pts/1    00:00:00 ps -ef
Techie@ravi:~/Tejaaa$
```

Kill a running process.

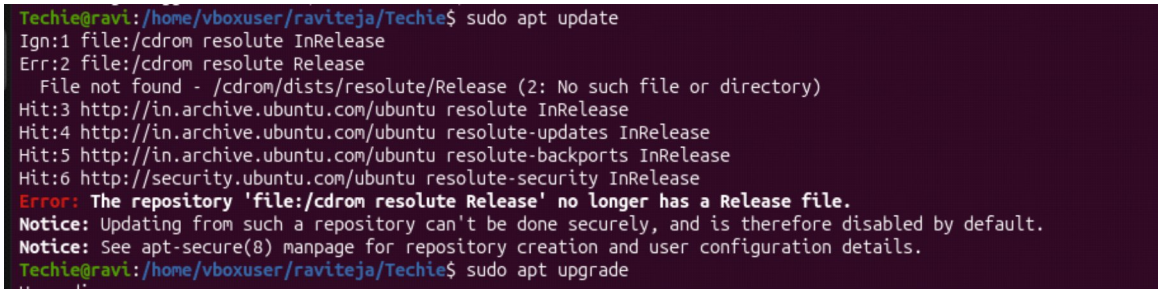
```
Techie@ravi:/home/vboxuser/raviteja/Techie$ sudo apt install nginx
Installing:
  nginx

Installing dependencies:
  nginx-common

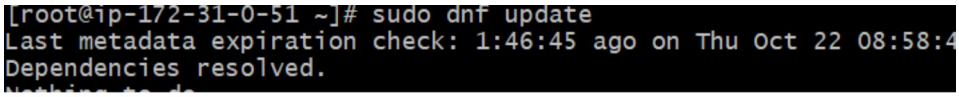
Suggested packages:
  fcgiwrap  nginx-doc

Summary:
```

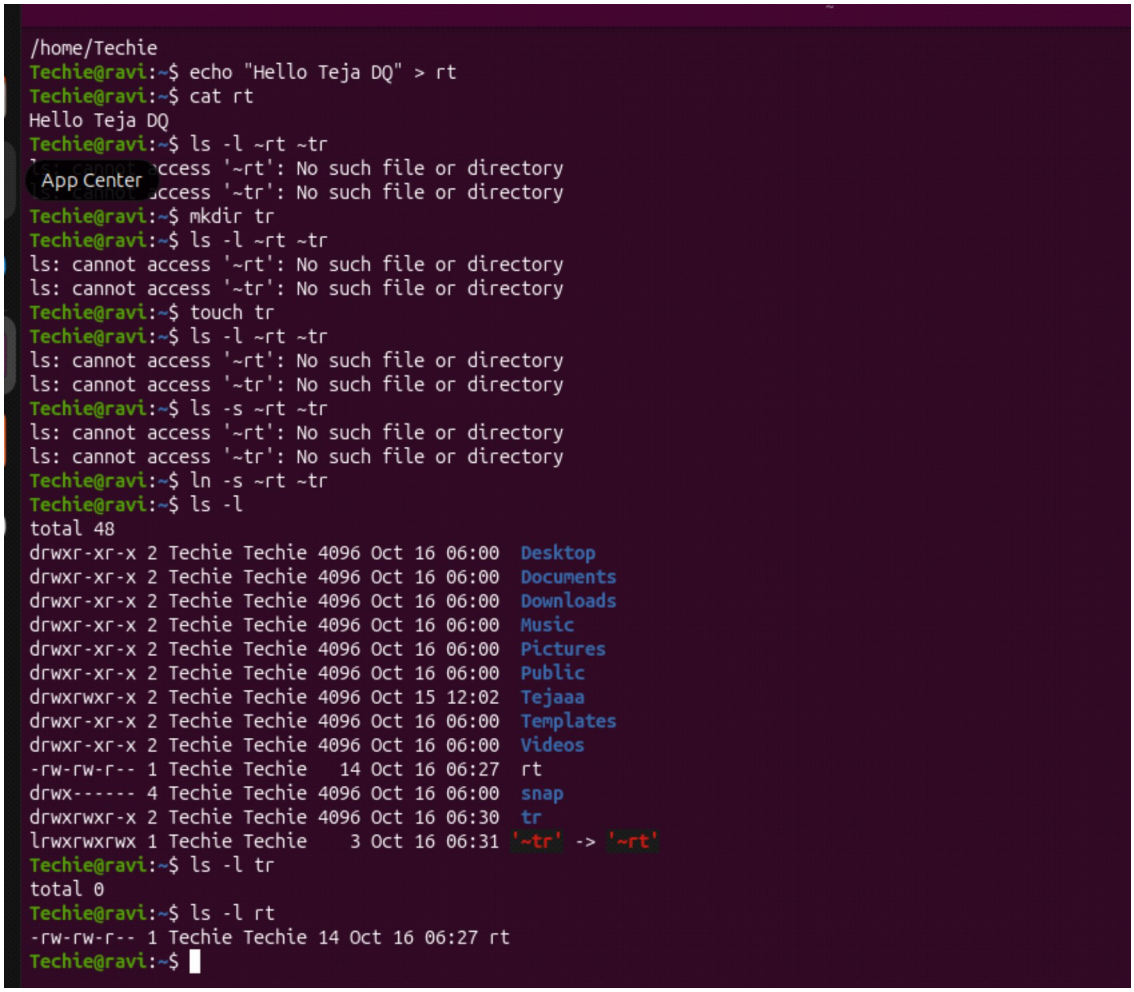
Install a package using the package manager (e.g., apt or yum).



Update the system packages.



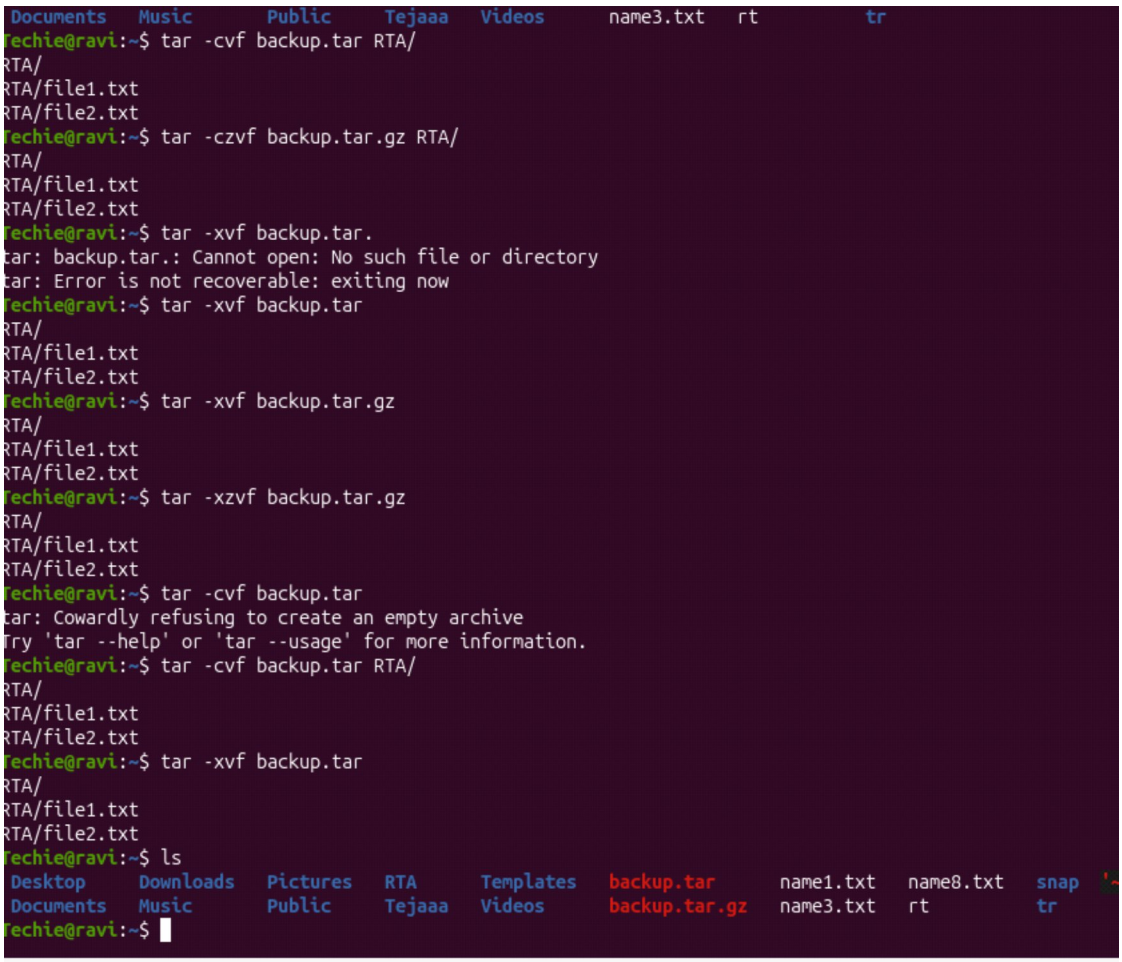
Create a symbolic link.



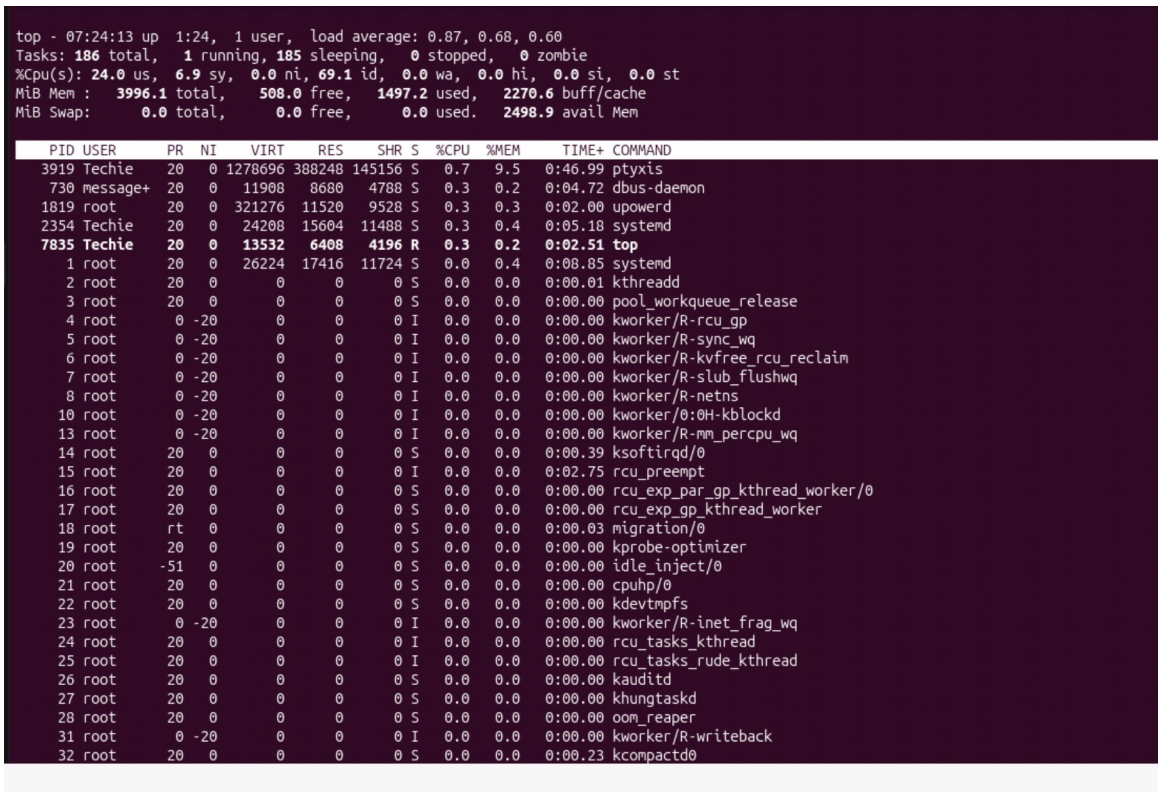
Search for files using the find command.

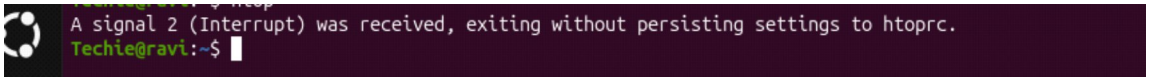


Compress and decompress files using tar.



Monitor system resources with top or htop.



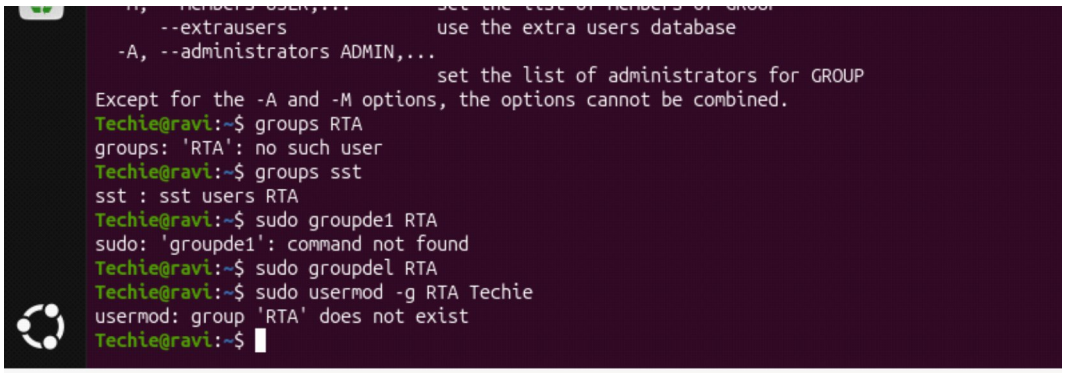


Create and manage user groups.

```
Techie@ravi:~$ touch sst
Techie@ravi:~$ sudo groupadd RTA
[sudo: authenticate] Password:
Techie@ravi:~$ group RTA
Command 'group' not found, did you mean:
  command 'groups' from deb coreutils-from-gnu (0.0.0~ubuntu25)
  command 'groups' from deb coreutils-from-uutils (0.0.0~ubuntu25)
  command 'groups' from deb coreutils-from-busybox (0.0.0~ubuntu25)
  command 'groups' from deb coreutils-from-toybox (0.0.0~ubuntu25)
Try: sudo apt install <deb name>
Techie@ravi:~$ getent group RTA
RTA:x:1002:
Techie@ravi:~$ cat /etc/group/
cat: /etc/group/: Not a directory
Techie@ravi:~$ cat /etc/group
root:x:0:
Welcome to Ubuntu
sys:x:3:
adm:x:4:syslog,vboxuser
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:
fax:x:21:
voice:x:22:
cdrom:x:24:vboxuser
floppy:x:25:
tape:x:26:
sudo:x:27:vboxuser,Techie
audio:x:29:
```

```
Techie@ravi:~$ groups Techie
Techie : Techie sudo users
Techie@ravi:~$ groups
Techie sudo users
Techie@ravi:~$ sudo usermod -aG RTA sst
usermod: user 'sst' does not exist
Techie@ravi:~$ sudo usermod -aG RTA Techie
Techie@ravi:~$ groups Techie
Techie : Techie sudo users RTA
Techie@ravi:~$ groups
Techie sudo users
Techie@ravi:~$ sudo adduser sst
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for sst
Enter the new value, or press ENTER for the default
        Full Name []: sst
        Room Number []: 27
        Work Phone []: 80195
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
Techie@ravi:~$ sudo usermod -aG RTA sst
Techie@ravi:~$ groups dev
groups: 'dev': no such user
Techie@ravi:~$ groups sst
sst : sst users RTA
Techie@ravi:~$ sudo gpasswg -d RTA
sudo: 'gpasswg': command not found
Techie@ravi:~$ sudo gpasswd -d RTA
Usage: gpasswd [option] GROUP

Options:
  -a, --add USER                add USER to GROUP
  -d, --delete USER             remove USER from GROUP
  -h, --help                    display this help message and exit
  -Q, --root CHROOT_DIR         directory to chroot into
  -r, --remove-password         remove the GROUP's password
```



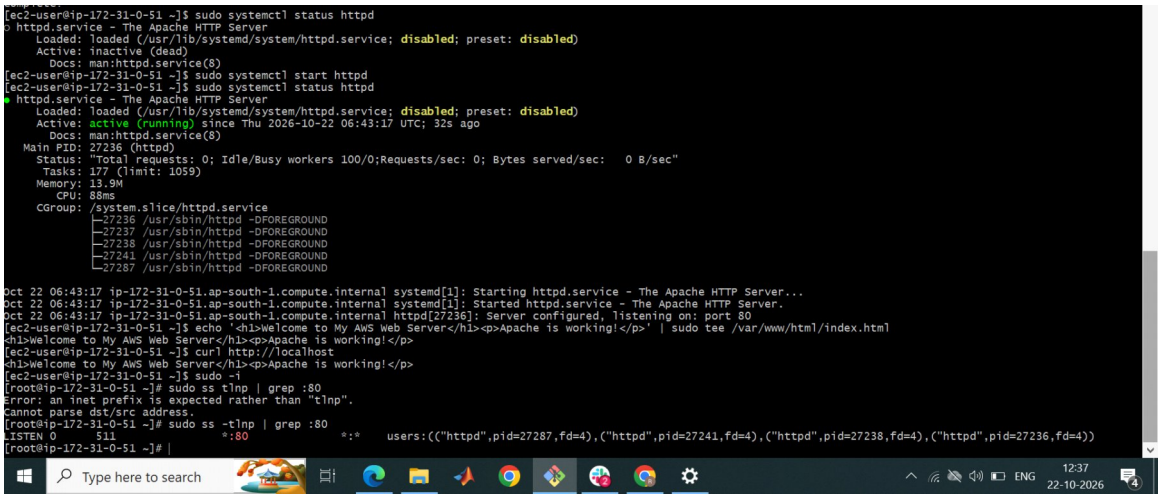
Set up SSH password less authentication.



Monitor log files using tail or grep.

```
-bash: $'\E[200~echo': command not found
[ec2-user@ip-172-31-0-51 ~]$ echo "ERROR: Database connection failed" >> app.log
[ec2-user@ip-172-31-0-51 ~]$ echo "Application running" >> app.log~
[ec2-user@ip-172-31-0-51 ~]$ ^C
[ec2-user@ip-172-31-0-51 ~]$ echo "Application started" > app.log
[ec2-user@ip-172-31-0-51 ~]$ echo "ERROR: Database connection failed" >> app.log
[ec2-user@ip-172-31-0-51 ~]$ echo "Application running" >> app.log
[ec2-user@ip-172-31-0-51 ~]$ tail /var/log/syslog
tail: cannot open '/var/log/syslog' for reading: No such file or directory
[ec2-user@ip-172-31-0-51 ~]$ touch app.log
[ec2-user@ip-172-31-0-51 ~]$ ls -l app.log
-rw-r--r--. 1 ec2-user ec2-user 74 Oct 22 06:23 app.log
[ec2-user@ip-172-31-0-51 ~]$ echo "Application started" >> app.log
[ec2-user@ip-172-31-0-51 ~]$ echo "ERROR: Database connection failed" >> app.log
[ec2-user@ip-172-31-0-51 ~]$ echo "Application running" >> app.log
[ec2-user@ip-172-31-0-51 ~]$ cat app.log
Application started
ERROR: Database connection failed
Application running
Application started
ERROR: Database connection failed
Application running
[ec2-user@ip-172-31-0-51 ~]$ tail -n 10 app.log
Application started
ERROR: Database connection failed
Application running
Application started
ERROR: Database connection failed
Application running
[ec2-user@ip-172-31-0-51 ~]$ tail -n 3 app.log
Application started
ERROR: Database connection failed
Application running
[ec2-user@ip-172-31-0-51 ~]$ tail -f app.log
Application started
ERROR: Database connection failed
Application running
Application started
ERROR: Database connection failed
Application running
^C
[ec2-user@ip-172-31-0-51 ~]$ grep -i app.log
error
^C
[ec2-user@ip-172-31-0-51 ~]$ grep "error" app.log
[ec2-user@ip-172-31-0-51 ~]$ grep -i "error" app.log
ERROR: Database connection failed
ERROR: Database connection failed
[ec2-user@ip-172-31-0-51 ~]$
```

Set up a web server (e.g., Apache or Nginx).



Configure and secure a MySQL Database.

```
make-dummy-cert             mariabackup                 mariadb-convert-table-format  mariadb-plugin               mariadb-waitpid
man                         mariadb                     mariadb-dump                  mariadb-secure-installation  mariadbd
man-recode                  mariadb-access              mariadb-dumpslow              mariadb-service-convert      mariadbd-multi
man.man-db                  mariadb-admin               mariadb-find-rows             mariadb-setpermission        mariadbd-safe
mandb                       mariadb-backup              mariadb-fix-extensions        mariadb-show                 mariadbd-safe-helper
manpath                     mariadb-binlog              mariadb-hotcopy               mariadb-slap                 matchpathcon
mapfile                     mariadb-check               mariadb-import                mariadb-tzinfo-to-sql
mapscrn                     mariadb-conv                mariadb-install-db            mariadb-upgrade
[root@ip-172-31-0-51 ~]# sudo mariadb
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 10
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE devops_db;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> CREATE USER 'devops_user'@'localhost'
    -> IDENTIFIED BY '123456';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON devops_db.*
    -> TO 'devops_user'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| devops_db          |
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
4 rows in set (0.000 sec)

MariaDB [(none)]> SHOW GRANTS FOR 'devops_user'@'localhost';
+-----------------------------------------------------------------------------------------------------------------+
| Grants for devops_user@localhost                                                                                |
+-----------------------------------------------------------------------------------------------------------------+
| GRANT USAGE ON *.* TO `devops_user`@`localhost` IDENTIFIED BY PASSWORD '*6BB4837EB74329105EE4568DDA7DC67ED2CA2AD9' |
| GRANT ALL PRIVILEGES ON `devops_db`.* TO `devops_user`@`localhost`                                              |
+-----------------------------------------------------------------------------------------------------------------+
2 rows in set (0.000 sec)

MariaDB [(none)]>
```

Set up a Application Server (e.g.,Apache Tomcat)

```
HTTP request sent, awaiting response... 200 OK
Length: 14488199 (14M) [application/x-gzip]
Saving to: 'apache-tomcat-10.1.61.tar.gz'

apache-tomcat-10.1.61.tar.gz              100%[=====================================================================================================>]

2026-10-22 09:16:33 (152 KB/s) - 'apache-tomcat-10.1.61.tar.gz' saved [14488199/14488199]

[root@ip-172-31-0-51 ~]# ls -lh apache-tomcat-10.1.61.tar.gz
-rw-r--r--. 1 root root 14M Oct  6 23:13 apache-tomcat-10.1.61.tar.gz
[root@ip-172-31-0-51 ~]# sudo tar -xzf /tmp/apache-tomcat-10.1.61.tar.gz -C /opt
[root@ip-172-31-0-51 ~]# sudo mv /opt/apache-tomcat-10.1.61 /opt/tomcat
[root@ip-172-31-0-51 ~]# ls -l /opt/tomcat/bin/startup.sh
-rwxr-x---. 1 tomcat tomcat 1904 Oct  6 22:56 /opt/tomcat/bin/startup.sh
[root@ip-172-31-0-51 ~]# sudo chown -R tomcat:tomcat /opt/tomcat
[root@ip-172-31-0-51 ~]# sudo chmod +x /opt/tomcat/bin/*.sh
[root@ip-172-31-0-51 ~]# sudo vi /etc/systemd/system/tomcat.service
[root@ip-172-31-0-51 ~]# readlink -f "$(which java)"
/usr/lib/jvm/java-17-amazon-corretto.x86_64/bin/java
[root@ip-172-31-0-51 ~]# sudo systemctl daemon-reload
[root@ip-172-31-0-51 ~]# sudo systemctl start tomcat
[root@ip-172-31-0-51 ~]# sudo systemctl status tomcat
● tomcat.service - Apache Tomcat Application Server
     Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: disabled)
     Active: active (running) since Thu 2026-10-22 09:22:30 UTC; 12s ago
    Process: 38092 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=0/SUCCESS)
   Main PID: 38099 (java)
      Tasks: 30 (limit: 1059)
     Memory: 79.9M
        CPU: 4.223s
     CGroup: /system.slice/tomcat.service
             └─38099 /usr/lib/jvm/java-17-amazon-corretto/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties

Oct 22 09:22:30 ip-172-31-0-51.ap-south-1.compute.internal systemd[1]: Starting tomcat.service - Apache Tomcat Application Server...
Oct 22 09:22:30 ip-172-31-0-51.ap-south-1.compute.internal startup.sh[38092]: Tomcat started.
Oct 22 09:22:30 ip-172-31-0-51.ap-south-1.compute.internal systemd[1]: Started tomcat.service - Apache Tomcat Application Server.

[root@ip-172-31-0-51 ~]# curl -I http://localhost:8080
HTTP/1.1 200
Content-Type: text/html;charset=UTF-8
Date: Thu, 22 Oct 2026 09:23:19 GMT

[root@ip-172-31-0-51 ~]# sudo tail -f /opt/tomcat/logs/catalina.out
22-Oct-2026 09:22:32.849 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application directory [/opt/
```

create a service file for Apache Tomcat.(Should execute by using systemtctl command)

```
[root@ip-172-31-0-51 ~]# sudo vi /etc/systemd/system/tomcat.service
[root@ip-172-31-0-51 ~]# readlink -f "$(which java)"
/usr/lib/jvm/java-17-amazon-corretto.x86_64/bin/java
[root@ip-172-31-0-51 ~]# sudo systemctl daemon-reload
[root@ip-172-31-0-51 ~]# sudo systemctl start tomcat
[root@ip-172-31-0-51 ~]# sudo systemctl status tomcat
● tomcat.service - Apache Tomcat Application Server
     Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: disabled)
     Active: active (running) since Thu 2026-10-22 09:22:30 UTC; 12s ago
    Process: 38092 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=0/SUCCESS)
   Main PID: 38099 (java)
      Tasks: 30 (limit: 1059)
     Memory: 79.9M
        CPU: 4.223s
     CGroup: /system.slice/tomcat.service
             └─38099 /usr/lib/jvm/java-17-amazon-corretto/bin/java -Djava.util.logging.config

Oct 22 09:22:30 ip-172-31-0-51.ap-south-1.compute.internal systemd[1]: Starting tomcat.service
Oct 22 09:22:30 ip-172-31-0-51.ap-south-1.compute.internal startup.sh[38092]: Tomcat started.
Oct 22 09:22:30 ip-172-31-0-51.ap-south-1.compute.internal systemd[1]: Started tomcat.service

[root@ip-172-31-0-51 ~]# curl -I http://localhost:8080
```

Print specific columns from a delimited file.

```
[root@ip-172-31-0-51 ~]# cut -d 'delimiter' -f field_number filename
cut: the delimiter must be a single character
Try 'cut --help' for more information.
[root@ip-172-31-0-51 ~]# cat > employees.csv <<'EOF'
ID,Name,Department,Salary
101,Ravi,IT,30000
102,Rahul,HR,25000
103,Sita,DevOps,40000
EOF
[root@ip-172-31-0-51 ~]# cut -d ',' -f 2 employees.csv
Name
Ravi
Rahul
Sita
```

Filter and print lines based on a specific pattern or condition.

```
[root@ip-172-31-0-51 ~]# touch filename
[root@ip-172-31-0-51 ~]# grep "pattern" filename
[root@ip-172-31-0-51 ~]# cat > app.log <<'EOF'
INFO Application started
INFO User logged in
ERROR Database connection failed
WARNING Disk space is low
ERROR Application crashed
INFO Application stopped
EOF
[root@ip-172-31-0-51 ~]# grep "ERROR" app.log
ERROR Database connection failed
ERROR Application crashed
[root@ip-172-31-0-51 ~]# sudo journalctl | grep -i "error"
Oct 22 06:14:18 ip-172-31-0-51.ap-south-1.compute.internal amazon-ssm-ag
```

Calculate and print the average, sum, or other statistics of a column.

```
[root@ip-172-31-0-51 ~]# cat > salaries.txt << 'EOF'
> ^[[200~Ravi 30000
> Rahul 25000
> Sita 40000
> Arun 35000
> EOF
[root@ip-172-31-0-51 ~]# cat salaries.txt
Ravi 30000
Rahul 25000
Sita 40000
Arun 35000
[root@ip-172-31-0-51 ~]# awk '{sum += $2} E
> ^C
[root@ip-172-31-0-51 ~]# awk '{sum += $2} END {print "Sum =", sum}' salaries.txt
Sum = 130000
[root@ip-172-31-0-51 ~]# awk END '{print "Sum =", sum}' salaries.txt
awk: cmd. line:1: END blocks must have an action part
[root@ip-172-31-0-51 ~]# awk '{sum += $2; count++} END {print "Average =", sum/count}' salaries.txt
Average = 32500
[root@ip-172-31-0-51 ~]# awk 'NR==1 || $2<min {min=$2} END {print "Minimum =", min}' salaries.txt
Minimum = 25000
[root@ip-172-31-0-51 ~]# awk 'END {print "Employees =", NR}' salaries.txt
Employees = 4
[root@ip-172-31-0-51 ~]#
```

Perform string manipulation, such as extracting substrings or changing case.

```
[root@ip-172-31-0-51 ~]# echo "hey Ravi" text.txt
hey Ravi text.txt
[root@ip-172-31-0-51 ~]# cat text.txt
cat: text.txt: No such file or directory
[root@ip-172-31-0-51 ~]# echo "hey Ravi" text.txt
hey Ravi text.txt
[root@ip-172-31-0-51 ~]# echo "hey Ravi" > text.txt
[root@ip-172-31-0-51 ~]# cat text.txt
hey Ravi
[root@ip-172-31-0-51 ~]# echo "hey Ravi" | cut -c 1-5
hey R
[root@ip-172-31-0-51 ~]# echo "hello linux" | tr 'a-z' 'A-Z'
HELLO LINUX
[root@ip-172-31-0-51 ~]# echo "text.txt" | tr 'a-z' 'A-Z'
TEXT.TXT
[root@ip-172-31-0-51 ~]# echo "Hello World" | sed 's/World/Linux/'
Hello Linux
[root@ip-172-31-0-51 ~]# echo "cat cat cat" | sed 's/cat/dog/g'
dog dog dog
[root@ip-172-31-0-51 ~]# echo -n "Hello Linux" | wc -c
11
[root@ip-172-31-0-51 ~]#
```

Count the occurrences of a specific pattern in a file.

```
[root@ip-172-31-0-51 ~]# cat app.log
INFO Application started
INFO User logged in
ERROR Database connection failed
WARNING Disk space is low
ERROR Application crashed
INFO Application stopped
[root@ip-172-31-0-51 ~]# grep -c "ERROR" app.log
2
[root@ip-172-31-0-51 ~]# grep -o "ERROR" app.log | wc -l
2
[root@ip-172-31-0-51 ~]# grep -oi "error" app.log | wc -l
2
[root@ip-172-31-0-51 ~]#
```

Sort lines based on a specific field or column.

```
[root@ip-172-31-0-51 ~]# cat > employees.txt <<'EOF'
Ravi IT 30000
Rahul HR 25000
Sita DevOps 40000
Arun IT 35000
EOF
[root@ip-172-31-0-51 ~]# cat employees.txt
Ravi IT 30000
Rahul HR 25000
Sita DevOps 40000
Arun IT 35000
[root@ip-172-31-0-51 ~]# sort -k1 employees.txt
Arun IT 35000
Rahul HR 25000
Ravi IT 30000
Sita DevOps 40000
[root@ip-172-31-0-51 ~]# sort -k3 -n employees.txt
Rahul HR 25000
Ravi IT 30000
Arun IT 35000
Sita DevOps 40000
[root@ip-172-31-0-51 ~]# sort -k3 -nr employees.txt
Sita DevOps 40000
Arun IT 35000
Ravi IT 30000
Rahul HR 25000
[root@ip-172-31-0-51 ~]#
```

Merge multiple files based on a common field or column.

```
[root@ip-172-31-0-51 ~]# cat > employees.txt <<'EOF'
101 Ravi
102 Rahul
103 Sita
104 Arun
EOF
[root@ip-172-31-0-51 ~]# cat > salaries.txt <<'EOF'
101 30000
102 25000
103 40000
105 35000
EOF
[root@ip-172-31-0-51 ~]# join employees.txt salaries.txt
101 Ravi 30000
102 Rahul 25000
103 Sita 40000
[root@ip-172-31-0-51 ~]# join -a 1 employees.txt salaries.txt
101 Ravi 30000
102 Rahul 25000
103 Sita 40000
104 Arun
[root@ip-172-31-0-51 ~]# join -a 1 -a 2 employees.txt salaries.t
101 Ravi 30000
102 Rahul 25000
103 Sita 40000
104 Arun
105 35000
[root@ip-172-31-0-51 ~]#
```

Substitute text in a file using search and replace.

```
[root@ip-172-31-0-51 ~]# cat > app.txt <<'EOF'
Hello World
Linux is easy
World is beautiful
Welcome to World
EOF
[root@ip-172-31-0-51 ~]# cat app.txt
Hello World
Linux is easy
World is beautiful
Welcome to World
[root@ip-172-31-0-51 ~]# sed 's/World/Linux/' app.txt
Hello Linux
Linux is easy
Linux is beautiful
Welcome to Linux
[root@ip-172-31-0-51 ~]#
```

Delete specific lines based on a pattern or line number.

```
[root@ip-172-31-0-51 ~]# cat > app.txt <<'EOF'
INFO Application started
ERROR Database connection failed
INFO User logged in
WARNING Disk space is low
ERROR Application crashed
INFO Application stopped
EOF
[root@ip-172-31-0-51 ~]# cat app.txt
INFO Application started
ERROR Database connection failed
INFO User logged in
WARNING Disk space is low
ERROR Application crashed
INFO Application stopped
[root@ip-172-31-0-51 ~]# sed '3d' app.txt
INFO Application started
ERROR Database connection failed
WARNING Disk space is low
ERROR Application crashed
INFO Application stopped
[root@ip-172-31-0-51 ~]#
```

Append or insert text before or after a specific pattern or line.

```
[root@ip-172-31-0-51 ~]# sed '2i\INFO Application initialization' app.txt
INFO Application started
INFO Application initialization
ERROR Database connection failed
INFO User logged in
WARNING Disk space is low
ERROR Application crashed
INFO Application stopped
[root@ip-172-31-0-51 ~]#
```

Print only specific lines from a file.

```
INFO Application stopped
[root@ip-172-31-0-51 ~]# sed -n '3p' app.txt
INFO User logged in
[root@ip-172-31-0-51 ~]#
```

Copy file from linux to windows machine

```
[ec2-user@ip-172-31-0-51 ~]$ ls
app.log  app.log~
[ec2-user@ip-172-31-0-51 ~]$ touch amazon.txt
[ec2-user@ip-172-31-0-51 ~]$ ls
amazon.txt  app.log  app.log~
[ec2-user@ip-172-31-0-51 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-0-51 ~]$ exit
logout
Connection to 13.232.133.47 closed.

hp1@DESKTOP-7S6NEQ6 MINGW64 ~/Downloads
$  scp -i key.pem ec2-user@13.232.133.47:/home/ec2-user/amazon.txt ~/Downloads/ravi.txt

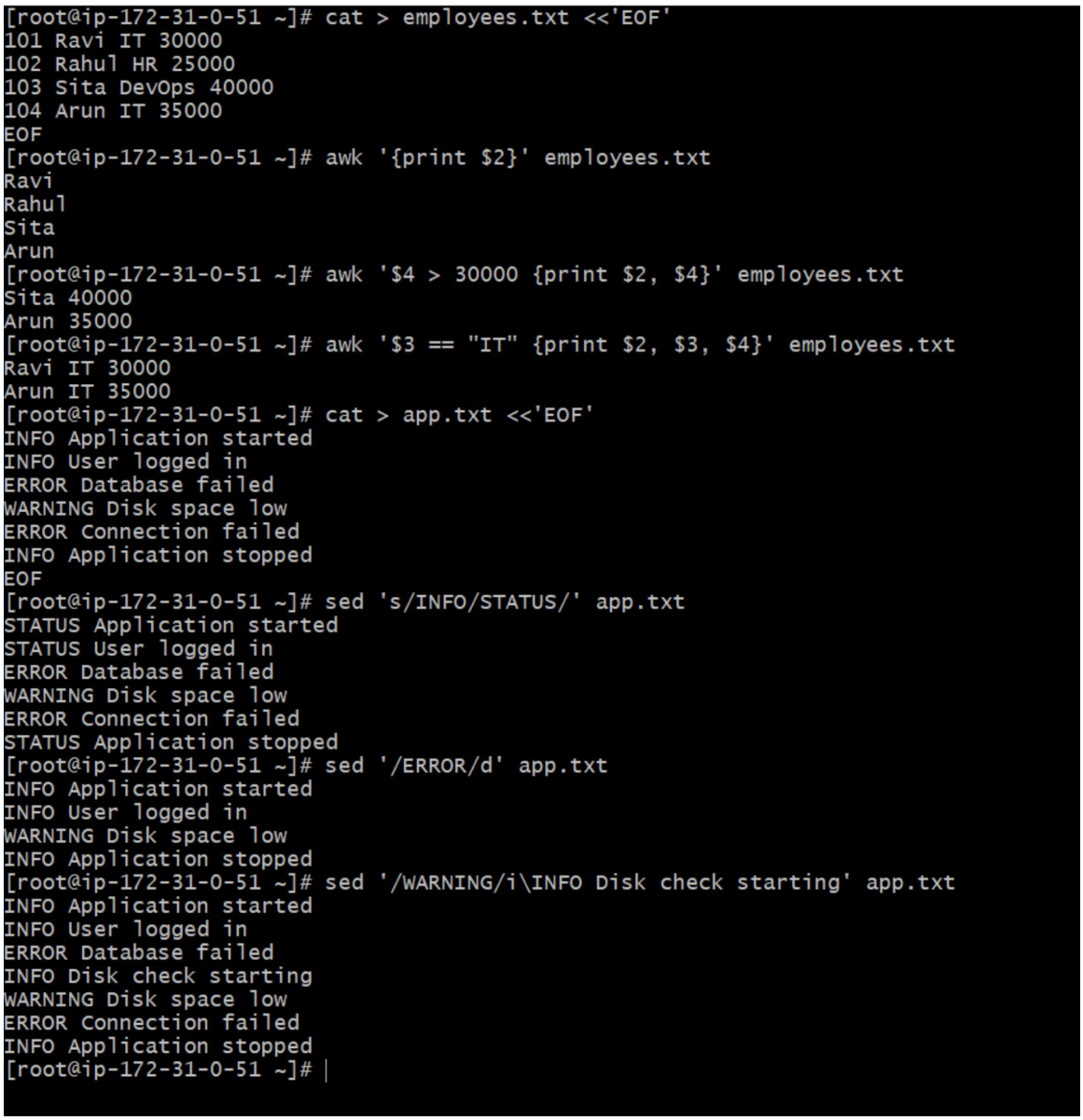
hp1@DESKTOP-7S6NEQ6 MINGW64 ~/Downloads
$  scp -i key.pem ec2-user@13.232.133.47:/home/ec2-user/amazon.txt ~/Downloads/amazon.txt

hp1@DESKTOP-7S6NEQ6 MINGW64 ~/Downloads
$  scp -i key.pem ec2-user@13.232.133.47:/home/ec2-user/amazon.txt ~/Downloads/ amazon.txt
amazon.txt: Not a directory

hp1@DESKTOP-7S6NEQ6 MINGW64 ~/Downloads
$  scp -i key.pem ec2-user@13.232.133.47:/home/ec2-user/amazon.txt ~/Downloads/amazon.txt

hp1@DESKTOP-7S6NEQ6 MINGW64 ~/Downloads
$
```

5 use cases for AWK and 5 use cases for sed


[root@ip-172-31-0-51 ~]# cat > employees.txt <<'EOF'
101 Ravi IT 30000
102 Rahul HR 25000
103 Sita DevOps 40000
104 Arun IT 35000
EOF
[root@ip-172-31-0-51 ~]# awk '{print $2}' employees.txt
Ravi
Rahul
Sita
Arun
[root@ip-172-31-0-51 ~]# awk '$4 > 30000 {print $2, $4}' employees.txt
Sita 40000
Arun 35000
[root@ip-172-31-0-51 ~]# awk '$3 == "IT" {print $2, $3, $4}' employees.txt
Ravi IT 30000
Arun IT 35000
[root@ip-172-31-0-51 ~]# cat > app.txt <<'EOF'
INFO Application started
INFO User logged in
ERROR Database failed
WARNING Disk space low
ERROR Connection failed
INFO Application stopped
EOF
[root@ip-172-31-0-51 ~]# sed 's/INFO/STATUS/' app.txt
STATUS Application started
STATUS User logged in
ERROR Database failed
WARNING Disk space low
ERROR Connection failed
STATUS Application stopped
[root@ip-172-31-0-51 ~]# sed '/ERROR/d' app.txt
INFO Application started
INFO User logged in
WARNING Disk space low
INFO Application stopped
[root@ip-172-31-0-51 ~]# sed '/WARNING/i\INFO Disk check starting' app.txt
INFO Application started
INFO User logged in
ERROR Database failed
INFO Disk check starting
WARNING Disk space low
ERROR Connection failed
INFO Application stopped
[root@ip-172-31-0-51 ~]#

A Raviteja